

# Maths

## Unit 2 - Numbers and Sequences

**WALT:** Use the relationship between repeated addition of a constant and multiplication to find and use a position to term rule.

**WALT:** Count on and count back in numbers of constant size, including fractions and decimals, and extend beyond zero to give negative numbers.



**With support, I  
can find the  
position to term  
rule for a  
sequence.**



**Independently,  
I can find the  
position to  
term rule for a  
sequence.**



**I can find the  
position to term  
rule for a more  
complicated  
sequence.**

**WALT:** Use the relationship between repeated addition of a constant and multiplication to find and use a position to term rule.

## Retrieval Practice – 3D shapes.

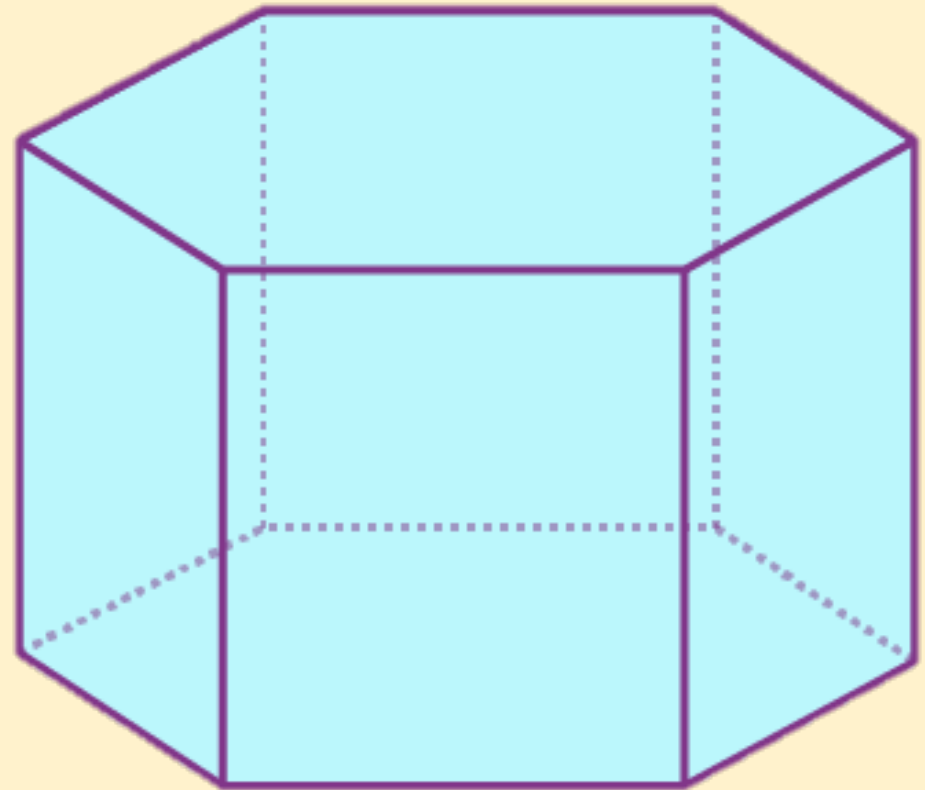
How many

Faces? – **8**

Edges? – **12**

Vertices? **8**

Does this 3D shape have?



**WALT:** Use the relationship between repeated addition of a constant and multiplication to find and use a position to term rule.

**Starter** – What is the term to term rule for this sequence?

**7, 14, 21, 28, 35**

**...**

**WALT:** Use the relationship between repeated addition of a constant and multiplication to find and use a position to term rule.

The **position to term** rule...

**explains how to find  
the value of any  
number in a sequence.**

**WALT:** Use the relationship between repeated addition of a constant and multiplication to find and use a position to term rule.

Let's find the **position to term** rule of the sequence from the starter.

**7, 14, 21, 28, 35 ...**

| Position | Term |
|----------|------|
| 1        | 7    |
| 2        | 14   |
| 3        | 21   |
| 4        | 28   |
| 5        | 35   |
| 6        |      |

**How could we find the 6<sup>th</sup> term in the sequence?**

**How could we find the 100<sup>th</sup> term in the sequence?**

**WALT:** Use the relationship between repeated addition of a constant and multiplication to find and use a position to term rule.

Let's find the **position to term** rule of the sequence from the starter.

**7, 14, 21, 28, 35 ...**

| Position | Term |
|----------|------|
| 1        | 7    |
| 2        | 14   |
| 3        | 21   |
| 4        | 28   |
| 5        | 35   |
| 6        |      |

What is the position to term rule for this sequence?

**Multiply the position by 7.**

**WALT:** Use the relationship between repeated addition of a constant and multiplication to find and use a position to term rule.

# Check your understanding

| Position | Term |
|----------|------|
| 1        | 11   |
| 2        | 22   |
| 3        | 33   |
| 4        | 44   |
| 5        | 55   |
| 6        | 66   |

What is the term to term rule for this sequence?

**Add 11**

What is the position to term rule for this sequence?

**Multiply the position by 11**

What is the 50<sup>th</sup> term in the sequence?

**550**

**WALT:** Use the relationship between repeated addition of a constant and multiplication to find and use a position to term rule.

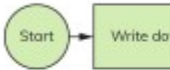
Exercise 2.1

1 a Find the position-to-term rule for the numbers in this table.

| Position | Term |
|----------|------|
| 1        |      |
| 2        |      |
| 3        |      |
| 4        |      |

b What is the 10th term?

- 2 The numbers in this sequence increase by equal amounts each time.
- a Write the three missing numbers.  
3, , , 15
- b What is the term-to-term rule for the sequence?
- c What is the position-to-term rule for the sequence?
- 3 a Follow the instructions.



b Jodi starts to make a table showing the position (shape number) and the term (total number of rectangles and triangles). Copy and complete her table.

| Position | Term |
|----------|------|
| 1        | 3    |
| 2        |      |
|          | 9    |
|          | 12   |
| 5        |      |

- c What is the position-to-term rule for Jodi's sequence?
- d What is the 50th term in the sequence?
- Check your answers with your partner.
- 5 Pablo counts up in quarters.
- What are the two missing numbers?
- $\frac{1}{4}$   $\frac{1}{2}$   $\frac{3}{4}$  1   $1\frac{1}{2}$   2
- 6 a Write a sequence with steps of constant size in which the first term is 1 and the fifth term is 1.04.
-     
- b What is the 10th term?
- 7 Ollie writes a number sequence starting at 15 and counting back in steps of 0.4.  
15, 14.6, 14.2, 13.8, ...
- He says, '1.5 cannot be in my sequence.'
- Ollie is correct. How do you know without counting back?
- Discuss your answer with your partner.
- 8 Hassan counts back in steps of  $\frac{2}{5}$  starting at 0.  
He counts 0,  $-\frac{2}{5}$ ,  $-\frac{4}{5}$ ,  $-\frac{6}{5}$ , ...
- Which of these numbers could Hassan say?
- $-\frac{4}{5}$   $-\frac{2}{5}$   $-\frac{6}{5}$   $-\frac{8}{5}$

9 Samira counts on from 20 in steps of 1.001

20    21.001    22.002    23.003 ....

Write the first number Samira says which is bigger than 30.

**WALT:** Use the relationship between repeated addition of a constant and multiplication to find and use a position to term rule.

- 1 a Find the position-to-term rule for the numbers in this table.

| Position | Term |
|----------|------|
| 1        | 6    |
| 2        | 12   |
| 3        | 18   |
| 4        | 24   |

**Multiply the position by 6**

- b What is the 10th term of the sequence 6, 12, 18 ...?

**60**

**WALT:** Use the relationship between repeated addition of a constant and multiplication to find and use a position to term rule.

**2** The numbers in this sequence increase by equal amounts each time.

**a** Write the three missing numbers.

3, 6, 9, 12, 15

**b** What is the term-to-term rule for the sequence? **Add 3**

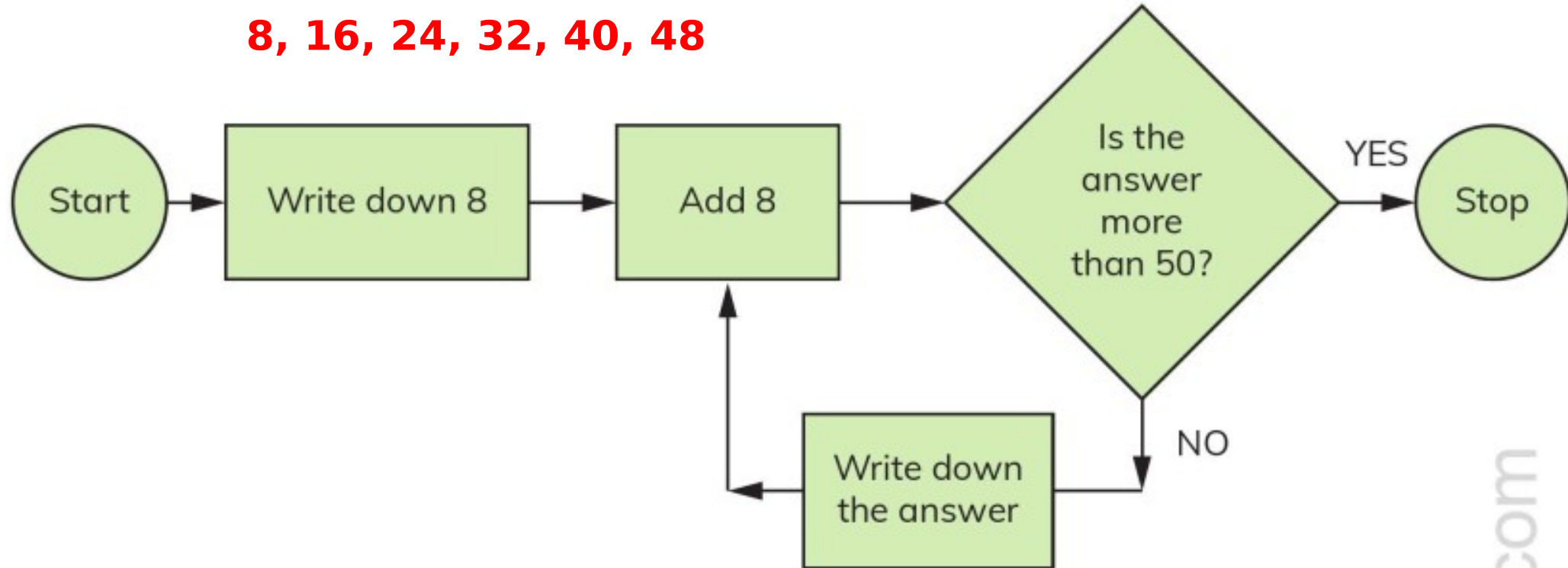
**c** What is the position-to-term rule for the sequence?

**Multiply the position by 3**

**WALT:** Use the relationship between repeated addition of a constant and multiplication to find and use a position to term rule.

3 a Follow the instructions in the flow diagram to generate a sequence.

**8, 16, 24, 32, 40, 48**



b What is the position-to-term rule for the sequence?

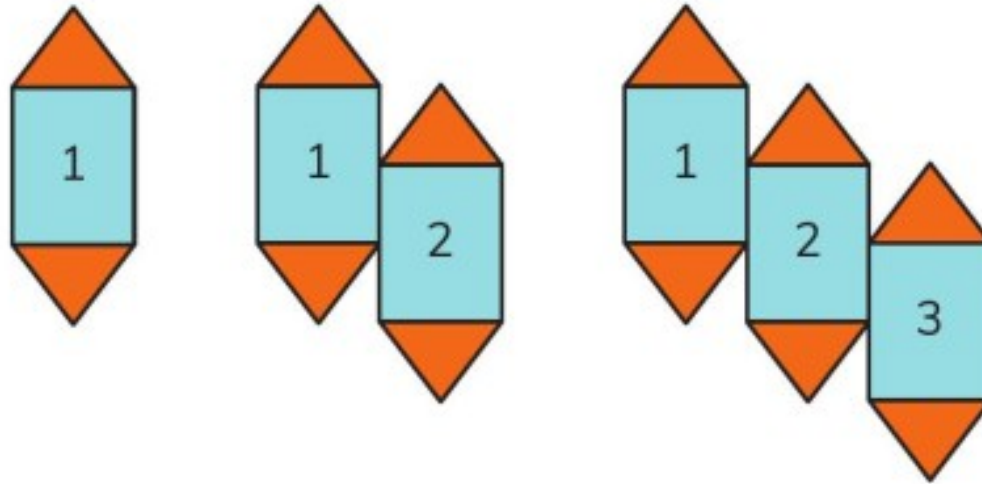
**Multiply the position by 8**

c Imagine the sequence continues forever.  
What is the 50th term in the sequence?

**400**

**WALT:** Use the relationship between repeated addition of a constant and multiplication to find and use a position to term rule.

- 4 Here is the start of a sequence using rectangles and triangles.  
Each rectangle is numbered.



The sequence continues in the same way.

- a How many triangles are there in the shape with 50 rectangles? **100**  
How many rectangles and triangles are there altogether in that shape?

**150**

**WALT:** Use the relationship between repeated addition of a constant and multiplication to find and use a position to term rule.

- b** Jodi starts to make a table showing the position (shape number) and the term (total number of rectangles and triangles). Copy and complete her table.

| Position | Term      |
|----------|-----------|
| 1        | 3         |
| 2        | <b>6</b>  |
| <b>3</b> | 9         |
| <b>4</b> | 12        |
| 5        | <b>15</b> |

- c** What is the position-to-term rule for Jodi's sequence?
- d** What is the 50th term in the sequence?

**Multiply the position by 3**

**150**

Check your answers with your partner.

**WALT:** Use the relationship between repeated addition of a constant and multiplication to find and use a position to term rule.

5 Pablo counts up in quarters.

What are the two missing numbers?

$\frac{1}{4}$     $\frac{1}{2}$     $\frac{3}{4}$    1    $1\frac{1}{4}$     $1\frac{1}{2}$     $1\frac{3}{4}$    2

6 a Write a sequence with steps of constant size in which the first term is 1 and the fifth term is 1.04.

1, 1.01, 1.02, 1.03, 1.04

b What is the 10th term? **1.09**

**WALT:** Use the relationship between repeated addition of a constant and multiplication to find and use a position to term rule.

7 Ollie writes a number sequence starting at 15 and counting back in steps of 0.4.

15, 14.6, 14.2, 13.8, ...

He says, '1.5 cannot be in my sequence.'

Ollie is correct. How do you know without counting back?

Discuss your answer with your partner.

7 All the tenths digits are even and 5 is odd.

**WALT:** Use the relationship between repeated addition of a constant and multiplication to find and use a position to term rule.

8 Hassan counts back in steps of  $\frac{2}{5}$  starting at 0.

He counts 0,  $-\frac{2}{5}$ ,  $-\frac{4}{5}$ ,  $-1\frac{1}{5}$ , ...

Which of these numbers could Hassan say?

$$-1-\frac{4}{5}$$

$$-2$$

$$-3$$

$$-3\frac{3}{5}$$

$$-4$$

**WALT:** Use the relationship between repeated addition of a constant and multiplication to find and use a position to term rule.



9 Samira counts on from 20 in steps of 1.001

20      21.001      22.002      23.003 ....

Write the first number Samira says which is bigger than 30.

**30.01**

**WALT:** Count on and count back in numbers of constant size, including fractions and decimals, and extend beyond zero to give negative numbers.

**Plenary** – What is the position to term rule in this sequence?

| Position | Term |
|----------|------|
| 1        | 10   |
| 2        | 16   |
| 3        | 22   |

**WALT:** Count on and count back in numbers of constant size, including fractions and decimals, and extend beyond zero to give negative numbers.



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